



Food storage, mobility, and the density-dependence of hunter-gatherer movement ecology

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ABSTRACT

Mobility, food storage, and population density are central to the movement ecology of hunter-gatherer populations and understanding how these lifestyle traits covary over time and space has long been of interest to archaeologists and anthropologists. An important question that remains unresolved is the conditions under which hunter-gatherer populations reduce the cost of mobility by increasing sedentism. Here, we model the interaction of movement frequency, distance, food storage technology and population density in ethnographic data. We show that increasing levels of food storage technology reduces annual movement frequencies but has little impact on annual total mobility costs: mobility costs are more often related to population density than storage capacity. This is because while food storage effectively increases the window of time over which resources are available, storage in non-food-producing economies cannot increase the net productivity of landscapes. Therefore, populations who move less frequently have to move further each time they move because resources remain finite and so become depleted. We derive a mathematical model of hunter-gatherer movement ecology based on optimization principles and scaling theory and test its predictions using spatially-explicit linear mixed models. We show that the interaction of mobility, storage, and population density in data are remarkably consistent with theoretical predictions. Our results suggest that while food storage is a technological response to seasonal environments, mobility reduction is primarily a behavioral response to increasing population density.

1. Introduction

Movement ecology focuses on understanding how, where, when, and why organisms move through space to make a living (Nathan et al., 2008). The movement ecology of hunter-gatherer societies has played a particular role in understanding the diversity of prehistoric lifestyles over time and space (Binford, 2001, 1980; Grove, 2010, 2009; Grove et al., 2023; Johnson, 2014; Kelly, 2013; Morgan, 2012; Testart et al., 1982; Whelan et al., 2013). This is because immediate access to harvestable food resources varies widely across the planet, mainly due to variation in the seasonal productivity of environments. As such, hunter-gatherer populations must make adaptive decisions of how, where, when, and why to move through space to encounter resources and conspecifics. In non-food-producing societies, adaptive responses to resource heterogeneity tradeoff mobility, sedentism, population density, technology, and storage (Binford, 1980; Kelly, 2013, 1992). Additionally, archaeologists have noted that many of the earliest agricultural economies seem to have been associated with semi-sedentary lifestyles, and so understanding the social, cultural, and environmental conditions

that promoted reduced mobility in subsistence societies has long been of interest (Baird et al., 2018; Bar-Yosef, 1998; Harris and Hillman, 1989; Moore et al., 2000; Scott, 2017). However, these lifestyle tradeoffs are complex and the archaeological record encodes only the material residues of these nuanced relationships. Ostensibly, archaeologists reverse engineer the patterning observed in the archaeological record through forensic reconstruction while relying on an archaeological record that is unavoidably under-determined, under-sampled, noisy, biased, and sparse. The behavioral drivers and evolutionary consequences of the material correlates we excavate do not preserve and so must be inferred from other data. Commonly in anthropological archaeology, this inference comes from analyzing behavioral models parameterized by the analysis of ethnographic data (Binford, 2001, 1980, 1978; Hamilton and Tallavaara, 2021; Kelly, 2013).

1.1. Ethnographic inference

Unlike ethnographic *analogy*, this form of ethnographic *inference* does not require the assumption that ethnographic societies are

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equivalent in some way to prehistoric societies (Kelly, 1999). Instead, consistent with statistical sampling theory and the central limit theorem, ethnographic inference simply assumes that a large-enough sample of ethnographic societies provides robust statistics that capture the macroscopic properties of a particular behavior or trait in question. Moreover, noisy estimates of ethnographic or ethnohistoric data pose no problem as long as the sample size is large enough and the estimates are not biased (Hamilton and Tallavaara, 2021). Given this, ethnographic inference assumes that the ethnographic record is the most reliable behavioral data of human foraging lifestyles and is therefore the primary source for building inferential behavioral models. Further, we assume the ethnographic data are a random sample (however noisy and imperfect) of the statistical population of hunter-gatherer lifestyles over time and space that we wish to make inferences about but can no longer observe directly. That is, rather than assuming any living hunter-gatherer population is somehow directly representative of prehistoric lifestyles—in some ways they may be, but in others, clearly not—we simply assume that the same kinds of social and environmental variables that cause the traits of populations to vary in the ethnographic present are likely to be fundamentally similar to those in the past, excluding the impact of modernization. In ethnographic data, we can estimate these statistics and model this variation.

In this paper, we ask the following question: when we observe variation in food storage and mobility in hunter-gatherers, what behavioral adaptive strategies are we actually observing? We wish to know under what conditions mobility is reduced and we want to know the consequences. We address this question by modeling ethnographic data and demonstrate that the interplay of mobility, storage and population density is more nuanced than commonly realized in the literature.

1.2. Food storage, population density, and mobility

We can think of annual mobility in hunter-gatherer populations as varying along two primary dimensions; the total distance traveled per year (km/yr), D , and the total number of moves made per year (events/yr), M . These lead to two important derivative measures of mobility; the average distance moved between patches (the ratio of total distance and movement frequency, km), $\langle D \rangle = D/M$, and patch residence time (inverse movement frequency, or the time between movement events, days/event), $T M^{-1}$. In subsistence-scale societies, mobility usually re-

fers to the relocation of individuals in space as local resources become depleted (Bettinger and Grote, 2016; Hamilton et al., 2016; Venkataraman et al., 2017) where movement decisions are modeled as trade-offs between the cumulative energy gains acquired while foraging in a patch versus the costs of relocating to another (Charnov, 1976). In this paper, we are particularly interested in understanding how storage technologies impact the movement ecology of hunter-gatherer societies across environments that vary widely in productivity and population density (Fig. 1).

There is a modularity to hunter-gatherer movement ecology as individuals often relocate as families, and discrete clusters of families constitute co-residential groups that fission and fuse through time (Hamilton et al., 2007b; Hill et al., 2011). Thus, at any one time, a hunter-gatherer population is comprised of multiple co-residential groups moving around their landscapes largely independently of each other, yet whose memberships are fluid and connected by a social interaction network. Groups are linked by fission–fusion dynamics at multiple scales, where neighboring co-residential groups may camp together at some parts of the year while disaggregating at others; families may choose to relocate individually or join other groups; spouses often relocate after marriage and individuals visit friends and relatives, moving from camp to camp. In this sense, hunter-gatherer populations are metapopulations of interconnected subpopulations (Hanski, 1999, 1998), where local interaction rates are high, but the network is bound by crucial weak ties created by individuals moving between groups (Apicella et al., 2012; Granovetter, 1973; Migliano et al., 2020). This social structure forms small-worlds (Buchanan et al., 2019; Hamilton, 2022). Thus, hunter-gatherer movement ecology is not only about encountering resources in space (Kaplan et al., 2000; Roberts and Stewart, 2018), but it is also about maintaining connectivity with conspecifics in the larger metapopulation (Dyble et al., 2016; Hamilton, 2022).

Ethnographic hunter-gatherer populations vary widely in their levels of residential mobility (Binford, 2001; Kelly, 2013; Smith and Coddington, 2021), from highly mobile groups who relocate camp every few days, such as the pre-contact Ache of northern Paraguay (Hill and Hurtado, 2017), to semi-sedentary groups who relocate villages on the scale of decades, such as those along the Northwest Coast of North America (Ames, 1994; Testart et al., 1982). This diversity is well-documented ethnographically, but also observed archaeologically. Moreover, the

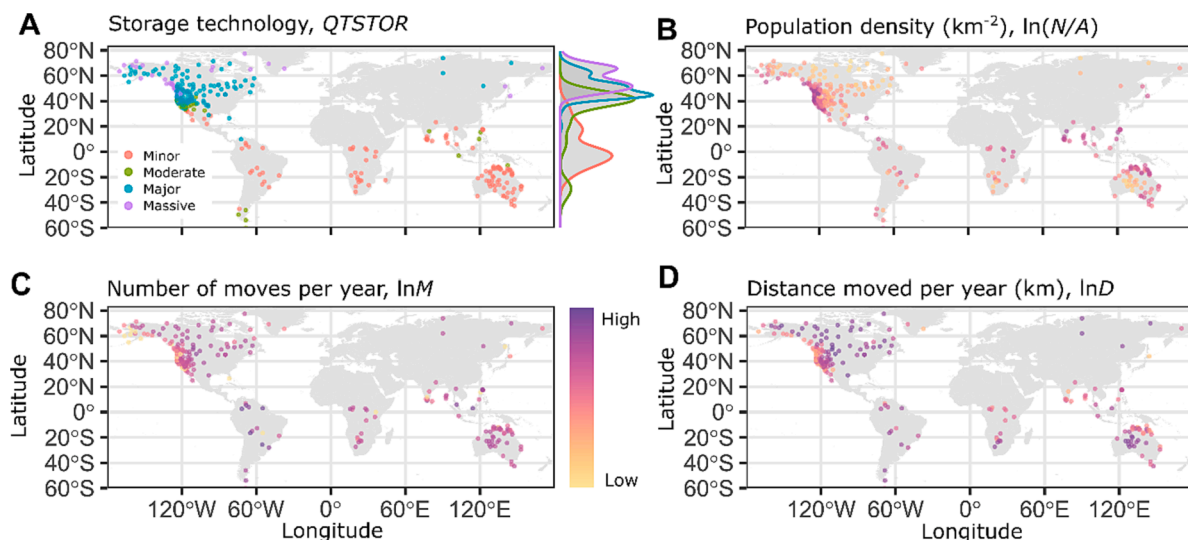


Fig. 1. The biogeographic distribution of storage technologies, population density, the number of moves per year, and the total distance moved per year in ethnohistoric hunter-gatherers using data from Binford (2001) ($n = 339$). A) Investment in storage, measured as Binford's (2001) QTSTOR an index of the quantity and duration of stored food resources, increases with latitude from minor investment at low latitudes to massive investment at high latitudes (vertical density plots on the right). B) Population densities tend to be highest along coasts and in the tropics; C and D) The biogeographic distribution of movement frequency and distance are similar, often low along coasts and high inland.

prevalence of sedentary lifestyles of prehistoric hunter-gatherers is becoming increasingly apparent raising the question of where, when, and why these lifestyles appear and evolve (Bailey and Spikins, 2008; Loftus et al., 2019; Marean, 2016, 2014).

1.3. Food storage and sedentism

Storage technology plays an important role in the biogeographic diversity of hunter-gatherer movement ecology (Fig. 1A), including both the quantity of food resources stored and the duration over which those stored resources are used (Binford, 2001; Kelly, 2013). Binford captures the quantity and duration of stored food resources in each society in his ethnohistoric sample by assigning them to one of four levels, $QTSTOR_{1-4}$: $QTSTOR_1$ is minor (or no) specialized storage technology, and so harvested food resources last no more than a few days; $QTSTOR_2$ is moderate investment in specialized storage technologies; $QTSTOR_3$ is major investment in anticipation of future use; and $QTSTOR_4$ is massive investment in the array of species stored and the duration of storage.

Methods of storing food resources may involve drying, smoking, freezing, pickling, or fermenting, depending on the resources available and the nature of the environment (Ames, 1994; Binford, 2001; Kelly, 2013; Morgan, 2012; Testart et al., 1982; Yoshikawa et al., 2022). Any of these methods may be employed in any of these categories of storage, though the degree and intensity of importance of stored food resources in the annual diet varies. In the higher categories, investment in storage technologies might involve the construction of specialized infrastructure, such as smoking/drying houses, storage facilities, or stone weirs necessary to capture the quantity of fish required to provide stored food resources for many hunter-gatherer populations of northern North America (for example Ames, 1994; Ames and Maschner, 1999; Balicki, 1989; Burch, 1998). Such intensive food storage may also involve large-scale, though often temporary aggregations of people in space to capture resources perhaps available only in a very limited window of opportunity in the year (Steward, 1938; Stewart et al., 2004). Further, food storage levels and methods are often correlated with institutions of resource ownership in hunter-gatherer societies (Freeman and Anderies, 2015).

Ethnographic observations of well-known semi-sedentary hunter-gatherer groups of the Northwest Coast of the Pacific Rim reflect lifestyles supported by storage technologies necessary to ameliorate the seasonal availability of key food resources, such as salmon (Thornton et al., 2015). Such lifestyles seem to suggest an inverse relationship between storage technologies and mobility. Similarly, highly mobile groups, such as the pre-contact Ache did not need to store food as they moved patches every few days (Hill and Hurtado, 2017). However, the degree to which this is a general pattern is unclear. Moreover, this simple model is often extrapolated into the archaeological record where evidence of prehistoric hunter-gatherer societies with storage technologies is assumed to suggest low mobility, and evidence of low mobility suggests high storage investment: this reasoning is engrained in discussions of the origins of agriculture where a necessary precondition for domestication and village lifestyles is assumed to be sedentism facilitated by storage (Bar-Yosef, 1998; Boyd, 2006; Scott, 2017).

Storage is also often associated with population density, where low storage capacity entails low population density, and high storage implies high population density. This follows from classic models of “population packing” (Keeley, 1988) or “circumscription”, where increased population density restricts hunter-gatherer mobility, forcing populations to move into marginal environments, increasing competition for food resources, leading to stress which reduces survival, stimulating technological innovations to solve ecological problems, such as storage, domestication, agriculture, and state formation (Binford, 2001; Carneiro, 1988, 1970, 1967; Cohen, 1995, 1977). In such scenarios, storage is a solution to demographic stress, and sedentism is exogenously driven. However, the biogeographic distributions of ethnohistoric populations shown in Fig. 1 suggests a more nuanced picture. Fig. 1A shows that

some groups exhibit massive investment in storage technology, but the biogeographic distribution of storage investment in general has a strong latitudinal signature, where hunter-gatherer groups at low latitudes ($< \pm 30^\circ$ latitude) commonly show minor investment in storage, with increasing storage investment toward the poles. This reflects the lower productivity and greater seasonality of climates at higher latitudes (Binford, 2001). Fig. 1B shows hunter-gatherer population densities tend to be high along coasts and in the tropics where storage is rare. However, throughout northern latitudes, population densities are generally low but storage technologies are of great importance. Fig. 1C and 1D show that mobility, both in terms of frequency of moves and total distance moved per year, are low along the Northwest Coast, for example, but the distribution of mobility and storage elsewhere around the world seems more tightly related to variation in population density than storage.

Storage is one adaptive dimension of the diversity of hunter-gatherer lifestyles while mobility and population density are separate dimensions. The question we are interested in here is how these different dimensions are related to each other. To do this, first, we derive a mathematical model of hunter-gatherer movement ecology based on principles of ecology and evolution using scaling theory, which generates testable predictions from data. Second, using ethnohistoric data from Binford (2001) we test these predictions using OLS regression and spatially-explicit linear mixed models. Third, we then discuss the implications of these results.

1.4. A mathematical model of hunter-gatherer movement ecology

From an evolutionary perspective, mobility is a pure cost as it takes time and energy to move through space (Alexander, 2003). As time and energy are finite resources this necessarily results in opportunity costs as resources spent moving cannot be allocated to other fitness-promoting tasks, such as foraging, reproduction, childcare, socializing, or tool manufacture, among others (Whelan et al., 2013). However, some resource expenditure on mobility is necessary to encounter scarce resources, perform daily tasks, and maintain contact with the broader social network of conspecifics. So, mobility costs can only be reduced to near zero under very particular circumstances. These opportunity costs lead to a straightforward optimization problem where mobility costs should be minimized to maximize the allocation of finite time and energy to other fitness-related pursuits.

To illustrate this optimization, we postulate an energy budget:

$$\begin{aligned} (\text{Total energy budget}) = & (\text{Somatic maintenance}) + (\text{Activity}) \\ & + (\text{Reproduction}) \end{aligned} \quad (1)$$

Here, the *Total energy budget* (i.e., the total calories foraged over some time period) is allocated to three fitness-promoting expenditures: *Somatic maintenance* is the energy required to maintain biological function; *Activity* is the energy required to support mobility; and *Reproduction* is the energy devoted to offspring. These three components are interdependent and constrained as any reduction in the total energy budget must be compensated by reducing one or more fitness-related expenditures. We can turn this energy budget model into a fitness-maximizing model by rearranging:

$$\begin{aligned} (\text{Reproduction}) = & (\text{Total energy budget}) - (\text{Somatic maintenance}) \\ & - (\text{Activity}) \end{aligned} \quad (2)$$

So, for a fixed total energy budget (i.e., the expected returns of a foraging strategy) and a fixed somatic maintenance cost (i.e., the combined basal metabolic rate of dependents), the energy allocated to reproduction is maximized by minimizing activity costs. Of course, activity costs involve the costs of performing all other necessary daily tasks (foraging, childcare, tool manufacture, camp relocation, and so on): the goal of the following formal model is then to understand how activity

costs should be minimized conditional on survival. We derive a model in two stages considering first the expected density-dependence of mobility, and then the effect of storage on density-dependent mobility.

1.5. Density-dependent mobility

Let B be the total energy budget of an individual, and R be the resource supply rate available from the environment; the ratio of the two is the home range of an individual, h_0 , or the foraging area required by an individual to meet their energy requirement (Burt, 1943; Jetz et al., 2004; Mace and Harvey, 1983; McNab, 1963):

$$h_0 = B/R \quad (3)$$

For a fixed energy budget and resource supply rate, h_0 is a constant in units of area and time. Empirically, R , can be approximated by net primary production, P , which is the flux of energy moving through an ecosystem in units of grams of carbon per unit time and space (Michaletz et al., 2017). The total area, A , required by a population of N individuals to meet their combined energy demand is then $A = \sum_1^N h_0$. Previous research (Hamilton et al., 2012, 2009, 2007a) shows that the relationship between hunter-gatherer population size, N , and territory size, A , (or approximately the total area used by a hunter-gatherer population over a year) across a sample of hunter-gatherer populations is well approximated by a power function of the form:

$$A = h_0 N^\beta \quad (4)$$

where β is a scaling exponent capturing the change in area resulting from a change in population size (or scale). If $\beta = 1$, then the scaling is linear and so the total population size is the linear sum of individual space requirements. However, empirically, $\beta \approx 3/4$, and significantly lower than 1 and so the territory size of a population is a sublinear function of population size (Hamilton et al., 2007). A more complete expression of territory size as a function of resource supply rate and population size is then

$$A = BR^{-1}N^{3/4} \quad (5)$$

Note that $A R^{-1}$, or, independent of population size, the territory size of a population increases linearly with decreasing resource supply rates from the environment. The area per individual is then

$$A/N = BR^{-1}N^{-1/4} \quad (6)$$

indicative of an economy of scale as the area an individual requires to meet their energy demand decreases with increasing population size (economies of scale refers to the lowering of cost per unit of output as the scale of activity increases). Strictly, equation (6) is the total amount of area an individual must access over the course of a year to meet their annual energy requirements, and so this is the sum of individual patches encountered over a year. If an individual moves an average of M times over the course of a year then the size of each patch, A_p , is approximately

$$A_p = \frac{A}{NM} = h_p N^{-1/4} \quad (7)$$

Where $h_p = BR^{-1}M^{-1}$, the area exclusively used by an individual in patch A_p to meet their resource requirements, or the proportion of the annual home range h_0 in the M th patch. If we then make the reasonable assumption that the cost of moving to a new patch is minimized (Hamilton et al., 2016), patches are contiguous and adjacent (Fig. 2) and so the average distance in units of km, $\langle D \rangle$, between two patches is proportional to the diameter of A_p and so

$$\langle D \rangle = 2 \left(\frac{A_p}{\pi} \right)^{1/2} \quad (8)$$

As $2\pi^{-1/2}$ is a constant, for convenience we set it to 1 and drop it from the following, and so $\langle D \rangle = A_p^{1/2}$.

Over the course of a year, an individual moves a total distance of D kilometers, moving an average distance of $\langle D \rangle$ kilometers M times, and so

$$D = \langle D \rangle M \quad (9)$$

If we then assume there is a well-defined average cost of moving per unit distance in space over the sample (i.e., kcals per kilometer), c_0 is a constant, then an individual's total energy cost of annual mobility is

$$C = c_0 D = c_0 \langle D \rangle M = c_0 \left(\frac{A}{N} \right)^{1/2} \quad (10)$$

in which case the annual cost of mobility is determined by the basic cost of movement, the average distance per move, and the number of moves, which increases with the linear dimension of the area an individual must access over a year to meet their energy requirements. Alternatively stated, the inverse of equation (10) states that the total cost of moving per year is density-dependent, described by a power law with exponent $-1/2$:

$$C = c_0 \left(\frac{N}{A} \right)^{-1/2} \quad (11)$$

Equation (11) shows that a doubling in population density results in a 50 % reduction of overall annual mobility costs. Holding all else constant, individuals in denser populations enjoy reduced mobility costs, not because they have less opportunities to move, but because it costs less to encounter resources and people in space.

1.6. The impact of storage on mobility

To consider the impact of storage on mobility, it is necessary to unpack the components of the total costs of annual mobility. Following equation (10), we can write

$$C = c_0 \langle D \rangle M = c_{(D)} \left(\frac{N}{A} \right)^{\beta_{(D)}} c_M \left(\frac{N}{A} \right)^{\beta_M} = c_D \left(\frac{N}{A} \right)^{\beta_D} \quad (12)$$

where $c_D = c_{(D)} c_M$ and $\beta_D = \beta_{(D)} + \beta_M$ (c_D has units of distance, and c_M has units of frequency). Because $D = \langle D \rangle M$ this then requires $\beta_{(D)} = \beta_M = \beta_D/2$, and as $\beta_D = -1/2$, then $\hat{\beta}_{(D)} = \hat{\beta}_M = -1/4$. We test these predictions below.

In the above derivation, the distance moved per year, D , is strictly a velocity as it is a linear displacement in one dimension of space, x , measured over some change in time, and so $D = \Delta x / \Delta t$, where Δt is one year, and Δx is measured in km. Similarly, M is a frequency as it is a number of events, E , measured over the same change in time, $M E / \Delta t$, and so $\langle D \rangle$ is simply a distance as $\langle D \rangle = D / M = \Delta x / E$ and so the units of Δt cancel. From this observation we can then derive an expression for patch residence time, T , as the inverse of the number of moves per year:

$$T = M^{-1} = \frac{\Delta t}{E} = c_M^{-1} \left(\frac{N}{A} \right)^{1/4} \quad (13)$$

Importantly, the constant c_M^{-1} now has units of time (the inverse of frequencies). Here, patch residence time is defined as the period between frequencies, or the change in time Δt with respect to a movement event, E , and equation (13) shows time increases with population density as a power law at a rate of $1/4$ (i.e., a doubling of population density leads to a 25 % increase in patch residence time).

In traditional conceptions of the marginal value theorem, patch residence time is a function of the diminishing returns of the cumulative gain function while foraging in a patch, and the decision to leave for a new patch is optimized at the point in which the marginal returns of the

gain function equal the environmental average (Charnov, 1976; Stephens and Krebs, 1986). Patch residence time is set by the overall resource availability in a patch, the rate at which those resources are depleted, and the known distance to a new patch (see Fig. 2). As such, hunter-gatherer food storage does not increase the net productivity of a patch, but it does decrease the rate at which resources are depleted by using technological innovations to increase effective availability of resources, extending the expected residence times. Thus, in our model, food storage decreases the frequency of mobility, by inflating the constant in equation (13), as is well-understood and well-explored in the literature. However, this is only half the story. Given the linear relationship of total distance and the number of moves per year shown in equation (9), increased patch residence time (or decreased frequency of movements) must be compensated for by increasing the average distance between patches, when those rarer moves are made. As a consequence, the total energy cost, C , set by the total distance moved per year, D , is predicted to be invariant to food storage because hunter-gatherer food storage technology primarily extends the availability of resources on the landscape, as opposed to increasing the productivity of environments; that is to say, our model suggests hunter-gatherer food storage technology does not increase environmental productivity per se, but

ameliorates fluctuating food resource availability in time and space.

This model leads to a number of testable predictions in data, three of which we explore here:

1.7. Predictions

Prediction 1: If ethnohistoric hunter-gatherers minimize the energy they expend on mobility then the total distance they move over a year as a function of population density should be a density-dependent power law with an exponent of -0.5 .

Prediction 2: If ethnohistoric hunter-gatherers minimize the energy they expend on mobility then the number of moves per year as a function of population density should be a density-dependent power law with an exponent of -0.25 .

Prediction 3: Food storage technology in ethnohistoric hunter-gatherers should decrease movement frequency (i.e., moves per year), but should not impact the total mobility costs (i.e., distance moved per year).

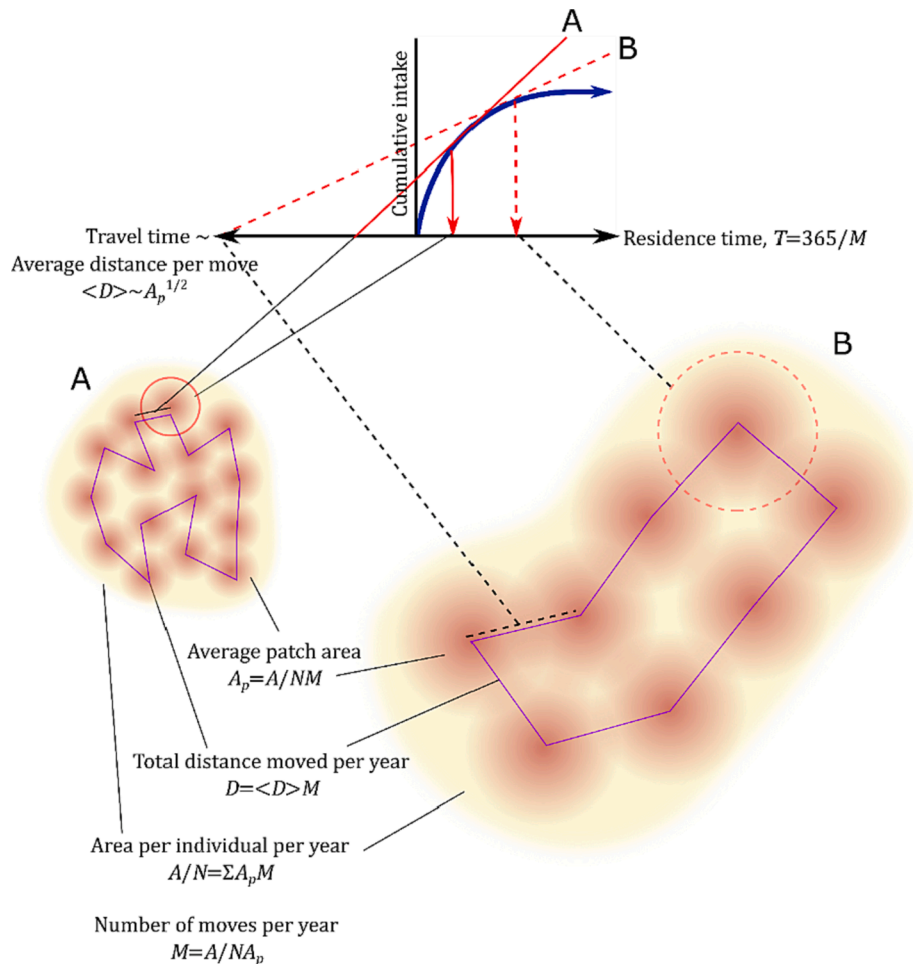


Fig. 2. The density-dependence of mobility illustrated in a schematic of the mobility model derived above. Hold environmental productivity constant and ignoring the effects of storage, here we see the annual mobility of average individuals in two populations, A and B. An individual in population A lives at a high density, patch sizes are small, and moves to new patches frequently. An individual in population B lives at a lower density, in much larger patches (and so inter-patch distances are greater), and moves between patches less frequently than an individual in population A. An ethnographic example of population A may be the Nukak of Columbia, and population B may be the Aka of Central Africa, both tropical foragers. In both cases, over the course of a year, individuals move from patch to patch (red circles) in order to meet their annual resource requirements (yellow patch). They move a total distance between patches over the year (purple line). The time spent in each patch (and therefore the density of patches to the area they need to access over a year) is determined by the gain function in the patch and the known distance to the next patch, as illustrated by the marginal value theorem shown at the top of the figure. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2. Methods

2.1. Data

To evaluate the predictions of the model we use data from Binford (2001). These are a sample of ethnohistoric hunter-gatherers coded in multiple ways. First, we excluded horse mounted foragers (Binford's SYSTATE = 1) from the dataset as, by definition, they have a fundamentally different movement ecology to pedestrian hunter-gatherers. Interestingly, however, we note we found no appreciable difference from the results we presented here when including equestrian groups.

We collected data on population density (continuous), N ; net primary production (continuous) measured in units of grams of carbon per square meter per year (i.e., annual carbon flux) extracted for each group from https://neo.gsfc.nasa.gov/view.php?datasetId=MOD17A3H_Y_NPP, P ; storage investment (ordinal), $QTSTOR$; total distance moved per year (continuous), D ; the number of moves per year (continuous), M ; and latitude and longitude (see SI files for data files). From Binford's total data set of 339 populations, our sample size is reduced to the 165 populations that have complete information on all the variables we include in this study. Following Binford (2001) the storage investment fixed effect, $QTSTOR$, has four levels (1–4): $QTSTOR_1$ is low or no investment in storage beyond a few days; $QTSTOR_2$ is moderate investment in storage, both quantity and duration; $QTSTOR_3$ is major investment in storage, both quantity and duration; and $QTSTOR_4$ is massive investment in storage, both quantity and duration.

Because linear mixed models do not produce standard measures of goodness of fit, such as R^2 , we use standard cross-validation methods. The data were divided into training data (70 %, $n = 127$) and testing data set (30 %, $n = 48$). Models were then run on the training data set and evaluated on their ability to predict the testing data set using K -fold cross validation. The goodness of fit was evaluated as cross validation R^2 where the ability of the model built on the training data is used to predict the testing data. All continuous data were transformed using the natural logarithm before analysis.

2.2. Statistical models

All statistical models were run in R version 4.2.2 statistical computing environment (R Development Core, 2009). Linear regression parameters were estimated using the lm function. Full models controlling for fixed and random effects used the *spaMM* package (Rousset, 2018; Rousset and Ferdy, 2014). *spaMM* analyzes spatially-explicit mixed models of the form $y = X\beta + Z\alpha + \epsilon$ where y is a vector of n dependent variables, X is a vector of independent variables, β is a vector of fixed effects regression coefficients, Z is the design matrix of the random effects, α is a vector of random effects regression coefficients, and ϵ is a vector of residuals not explained by the model. In these models, space (latitude and longitude) is modeled as a random effect. The exact forms of the fitted models were:

$$\ln Y(N/A, Q, Z|\beta) = c_0 + \beta_1 \ln\left(\frac{N}{A}\right) + \beta_2 QTSTOR_i + \beta_3 \left(\ln\left(\frac{N}{A}\right) \times QTSTOR_i\right) + \alpha Z + \epsilon \quad (14)$$

Where Y is a measure of mobility, N/A is population density, $QTSTOR_i$ is the fixed effect of storage investment in the i th level of storage technology.

3. Results

Fig. 3 shows boxplots of the dataset by storage technology. In all cases, distributions of these variables on the log scale are closer to normal than on the linear scale. Fig. 3A and 3B show that across the sample, neither the distance moved per year nor the number of moves per year differ dramatically across levels of storage investment. Fig. 3C shows there is more variation in average population density across the levels of storage investment. These plots indicate, on average, higher levels of storage are associated with low population density.

Fig. 4A shows log–log bivariate plots of the distance traveled per year as a function of the number of moves per year. OLS regression slopes are reported in the inset, and the regression statistics for each model are given in Table 1. Here, the total distance traveled per year is approximately linear with the number of moves per year, perhaps slightly superlinear where the total distance traveled per year increases by $\sim 110\%$ with every doubling of mobility frequency. Fig. 4B shows that hunter-gatherer population densities increase approximately linearly with environmental productivity (see Hamilton et al., 2020) and so more productive environments tend to support greater population densities. Fig. 4C shows that hunter-gatherer territory sizes scale sublinearly with population size, resulting in an economy of scale.

Fig. 5A shows that the number of moves per year decreases with population density as -0.27 ± 0.05 , close to the theoretically-predicted value of -0.25 . Fig. 5B shows the scaling of the total distance moved per year, D decreases as a function of population density, N , at a rate of -0.52 ± 0.06 , close to the theoretically-predicted value of -0.5 . Similarly, Fig. 5C shows the average distance per move is density dependent with an exponent -0.26 ± 0.04 , again, close to the theoretically predicted value of -0.25 . The results shown in Fig. 5A–C and Table 1 (models D–F) provide strong initial support for the hypotheses derived from the mathematical model, suggesting that hunter-gatherers minimize mobility costs by adapting mobility expenditures to local population densities.

Table 3 show the results of the equivalent spatial mixed models of hunter-gatherer annual mobility as those presented in Fig. 5 and Table 2. Model 1 shows that the moves per year decreases with population density with a slope of -0.19 ± 0.07 , slightly shallower than in the OLS scaling shown in Fig. 5D and Table 2, but consistent with the predicted value of -0.25 . The fixed effect of the level of storage has a statistically

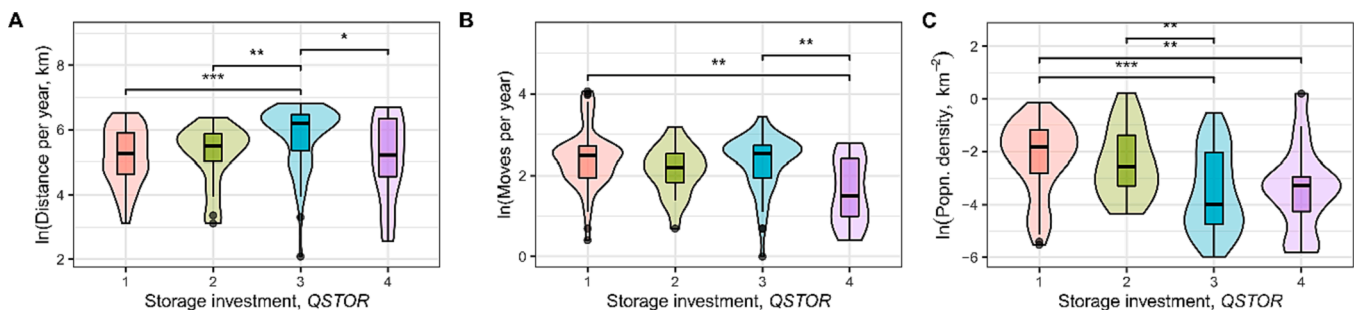


Fig. 3. Boxplots of the distance traveled per year (A), the number of moves per year (B), and population density (C) for ethnohistoric hunter-gatherer populations grouped by their investment in storage technology ($QTSTOR_{1-4}$ 1–4). Cross-bars are significant t -tests (95 % confidence) between groups, controlling for multiple comparisons: * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$.

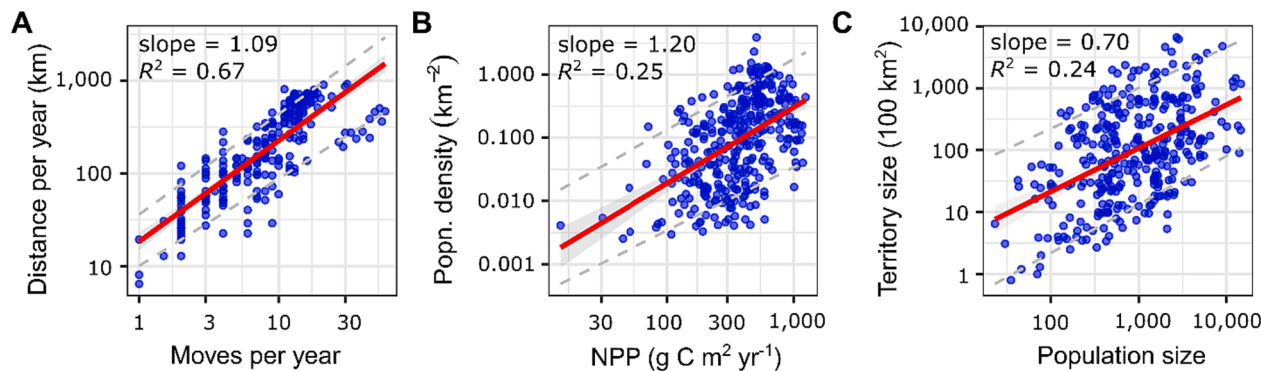


Fig. 4. OLS bivariate scaling of A) the distance moved per year by the number of moves per year, and B) population density as a function of environmental productivity. Red lines are OLS fits with 95% confidence intervals (light grey zone, and grey dashed lines are 10th and 90th quantiles of the scaling relationship (summary statistics of the models are given in Table 1). A) The total distance traveled per year is approximately a linear function of the number of moves per year. B) Hunter-gatherer population densities increase approximately linearly with environmental productivity. C) Hunter-gatherer territory sizes increase sublinearly with population sizes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Scaling statistics of mobility, population density, and territory size.

	Dependent variable:		
	Distance per year, lnD	Population density, lnN/A	Territory size, ln A
Moves per year, lnM	(Panel A) 1.09*** (0.99, 1.19)	(Panel B) –	(Panel C) –
Net primary production, lnNPP	–	1.20*** (0.98, 1.42)	–
Population size, lnN	–	–	0.70*** (0.57, 0.84)
Intercept	2.91*** (2.69, 3.13)	–9.49*** (–10.80, –8.19)	4.44*** (3.53, 5.36)
Observations	236	339	339
R ²	0.67	0.25	0.24
Residual Std. Error	0.62 (df = 234)	1.39 (df = 337)	1.57 (df = 337)
F Statistic	469.58*** (df = 1; 234)	114.10*** (df = 1; 337)	105.80*** (df = 1; 337)

Note: *p < 0.1; **p < 0.05; ***p < 0.01. Numbers in parentheses below the parameter estimates are the upper and lower bounds of the 95% C.I.

significant negative effect on movement frequency at all levels of storage intensity. The model has an $R^2 = 0.41$.

Model 2 shows that the total distance moved per year decreases with population density at a rate -0.47 ± 0.07 , very close to the rate predicted by the theory derived above. However, unlike the model of

movement frequency, the level of storage technology has no significant impact on the total distance moved per year, except for a significant negative effect in societies with massive storage investment. That is to say, among this sample of ethnohistoric hunter-gatherer populations, the total cost of mobility is density-dependent, but variation in storage

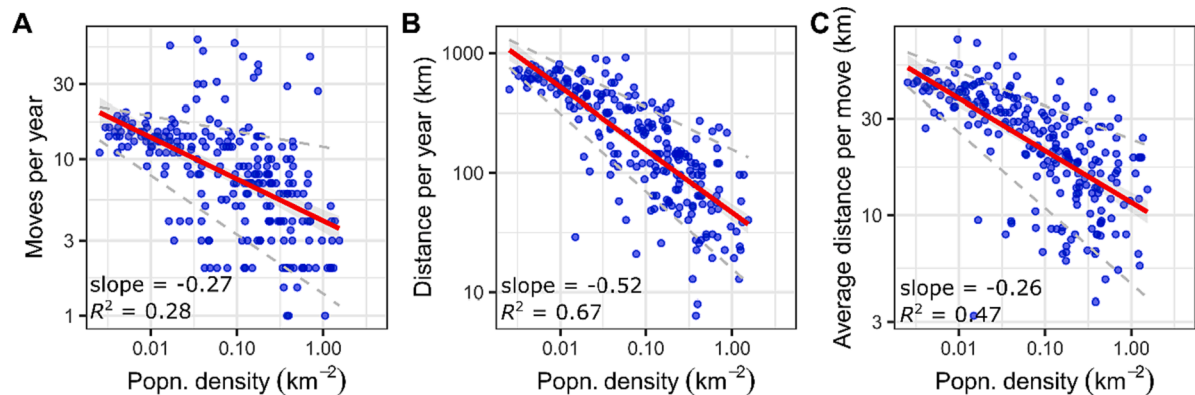


Fig. 5. OLS bivariate scaling of the three dimensions of hunter-gatherer mobility on logged axes. Red lines are OLS regression models with 95% confidence interval (light grey zone) (summary statistics of each model are given in Table 2) and the grey dashed lines are the 10th and 90th quantiles capturing the upper and lower bounds of the scaling relationships. A) The number of moves per year decreases with increasing population density with a slope of -0.27 not significantly different from the rate predicted by theory, -0.25 . B) The distance traveled per year decreases with population density with a slope of -0.52 , not significantly different from the rate predicted by theory, -0.5 . C) The average distance per move decreases with population density with a slope of -0.26 , not significantly different from the rate predicted by theory, -0.25 . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Scaling statistics of density-dependent annual mobility.

	Dependent variable:		
	Moves per year ln M	Distance per year ln D	Distance per move (lnD)
Popn. density, lnN/A	(Panel A) −0.27*** (−0.32, −0.21)	(Panel B) −0.52*** (−0.58, −0.47)	(Panel C) −0.26*** (−0.29, −0.22)
Intercept	1.39*** (1.23, 1.55)	3.84*** (3.68, 4.00)	2.45*** (2.34, 2.56)
Observations	236	236	236
R ²	0.28	0.61	0.43
Residual Std. Error (df = 234)	0.69	0.67	0.47
F Statistic (df = 1; 234)	92.90***	368.38***	178.10***

Note: *p < 0.1; **p < 0.05; ***p < 0.01. Numbers in parentheses below the parameter estimates are the upper and lower bounds of the 95% C.I.

technology has little impact. The model has an $R^2 = 0.68$. Thus, mobility costs are density dependent, but are largely independent of storage.

Model 3 shows that the average distance per move decreases with population density with a slope -0.29 ± 0.05 , consistent with the theoretical prediction of -0.25 . Average distance per move decreases at moderate levels of level of storage technology, but there is no effect in the major and massive investment categories. Thus, inter-patch movement distances are density-dependent, and not storage-dependent in societies with the higher levels of storage investment.

Thus, all the mobility parameters we observe in ethnographic data are consistent with our theoretical predictions derived from our model of hunter-gatherer movement ecology. These results also show that the inherent noise in this database, and others like it, is i.i.d. (independent and identically distributed) such that the signal can be extracted confidently from the noise (Hamilton and Tallavaara, 2021).

4. Discussion

Our results show that mobility costs in hunter-gatherer populations are density-dependent because in denser populations it is less expensive to encounter people and resources in space. While storage technology decreases the number of times people move per year it does not decrease the overall distance people have to move over the course of a year and so the total cost of mobility is unaffected by storage technology, except from in the highest storage investment societies. Our mathematical model of hunter-gatherer movement ecology predicts mobility costs should be minimized as time and energy spent moving cannot be allocated to other fitness-promoting behaviors. Testing the model with ethnographic data leads to four main results:

1) *Hunter-gatherers move less total distance over a year in denser populations because they don't have to access as much space to meet their energy budget.*

Given two independent measures of mobility in an ethnohistoric sample of 165 hunter-gatherer societies—the number of moves per year and the total distance traveled per year—this model led to three predictions. The first prediction is that the annual cost of mobility in hunter-gatherer societies is minimized, and this is tested showing that the total distance moved per year is negatively density-dependent, where the rate of density-dependence reflects the decreased amount of space individuals require to meet their energy demand with increasing population size. This prediction is well-supported by data.

2) *Hunter-gatherers move less frequently in denser populations, but each time they move, they must move further between patches.*

This second result shows that the number of moves per year is also negatively density-dependent, but at half the rate. This is because as the total distance moved per year is approximately a linear function of the number of moves per year, thus the average distance per move is con-

Table 3
Annual mobility spatial mixed model results.

	Dependent variable:		
	Moves per year lnM	Distance per year ln D	Distance per move (lnD)
Constant,	(Model 1) −2.12 (1.68, 2.56)	(Model 2) 4.17 (3.84, 4.50)	(Model 3) −2.06 (1.77, 2.34)
Population density, ln(N/A)	−0.19 (−0.33, −0.05)	−0.47 (−0.60, −0.34)	−0.29 (−0.39, −0.18)
Moderate storage, QTSTOR ₂	−1.04 (−1.77, −0.31)	−0.12 (−0.91, 0.67)	0.57 (0.04, 1.11)
Major storage, QTSTOR ₃	−1.19 (−1.92, −0.45)	−0.37 (−0.92, 0.18)	0.51 (−0.01, 1.03)
Massive storage, QTSTOR ₄	−1.67 (−2.51, −0.84)	−1.23 (−2.04, −0.42)	0.49 (−0.09, 1.08)
QTSTOR ₂ :ln(N/A)	−0.15 (−0.43, 0.13)	0.02 (−0.26, 0.31)	0.09 (−0.12, 0.29)
QTSTOR ₃ :ln(N/A)	−0.14 (−0.33, 0.06)	−0.06 (−0.24, 0.11)	0.05 (−0.09, 0.20)
QTSTOR ₄ :ln(N/A)	0.14 (−0.38, 0.11)	−0.14 (−0.38, 0.11)	0.01 (−0.17, 0.19)
Random effects			
v	16.67	3.36	16.67
ρ	0.27	7.05	0.63
Variance parameters			
λ = Var(u)	0.14	0.36	0.05
Residual variance			
Φ	0.27	0.02	0.14
Observations	120	120	120
Log Likelihood	−104.58	−110.95	−64.97
Testing R ²	0.41	0.68	0.64

Bolded values denote statistical significance determined by whether the 95 % confidence intervals (in parentheses) encompass 0.

stant with respect to movement frequency; i.e., if $D \propto M^1$ then $\langle D \rangle = D/M = M^0$. This then predicts that the two aspects of mobility that comprise the total annual cost of moving—moves per year and the average distance per move—are also density-dependent, but inversely related. This prediction is also well-supported by data.

- 3) *Independent of population density, storage technology reduces movement frequency, but does not reduce the total distance moved per year in most hunter-gatherer groups.*

Given results 1 and 2, we examined the interaction of storage investments on mobility costs. To do this we introduced measures of storage investment as fixed factors in a spatially-explicit mixed model of hunter-gatherer movement ecology that also included population density and spatial autocorrelation as predictors of mobility. Results show that increasing intensities of storage technology decreased movement frequency, but storage technology did not affect the total distance traveled per year in most groups. However, it is interesting to note that in this sample, massive investment in storage technology reduces both movement frequency and the total distance moved per year (Table 3).

- 4) *The density-dependence of mobility not only minimizes the costs of encountering resources, but it also minimizes the costs of maintaining connectivity with the metapopulation.*

These results give a more nuanced picture of hunter-gatherer movement ecology than generally assumed. For example, in an influential paper, Keeley suggests a robust and positive correlation among storage dependence, sedentism, and population density among ethnographic hunter-gatherers, where storage is influenced by population pressure, concluding population pressure is a “necessary and sufficient condition for and the efficient cause of complexity among hunter-gatherers” (Keeley, 1988). The model and the results we present here suggest that while population density and storage indeed correlate with the number of moves made per year, the total energy and time budget devoted to mobility remain unaffected in most groups; true hunter-gatherer sedentism would require both a reduction in the frequency of moves *and* the total distance moved per year, as observed only in ethnographic groups that show massive investment in storage technology. Importantly, our model suggests mobility reduction in the ethnohistoric record is not the result of exogenous constraints, such as population pressure restricting movement, but because more productive environments tend to support more people per unit area, denser populations require less space to meet their energy budget and so people can afford to reduce their mobility costs as they don’t have to access as much area to meet their resource demands. As a consequence, both mobility costs and mobility frequencies are endogenous and adaptively density-dependent.

We also show that while denser hunter-gatherer populations spend longer time in a patch supplemented by storage (sensu Binford, 1990, 1978), when they move, they have to move greater distances. Fig. 1A shows the latitudinal gradient of storage investment in hunter-gatherer biogeography, where high latitude populations have the most investment in storage technologies; not only does environmental productivity, and thus resource availability, decrease at higher latitudes, but climates are increasingly seasonal. This biogeographic pattern suggests that hunter-gatherer storage technology is an adaptation to increasingly seasonal environments, where resource availability is restricted in time and space and so resources must be stored for more extended periods and in more significant quantities. Additionally, storage is increasingly feasible in more northerly environments, due to decreased temperature-dependent decay rates and colder temperatures. Storage is not feasible in high productivity, tropical environments, for example, so access to depleting resources can only be through mobility (Politis, 2009, 1996). However, in these environments, the distances between patches are

reduced, and mobility frequency is high (Hamilton et al., 2016; Hill and Hurtado, 2017; Politis, 1996).

Hunter-gatherer movement ecology is not solely determined by access to food resources. Over the course of a year, hunter-gatherers must also maintain connections with the broader metapopulation, and this is achieved through a social network connected by a fluid fission–fusion interaction structure that also entails the costs of moving through space (Hamilton, 2022). Connectivity between subpopulations is essential to maintain the flow of genetic and cultural information necessary to promote long-term population viability and stability (Brown and Kodric-Brown, 1977; Hanski, 1999, 1998). It is straightforward to observe that the average distance between any two individuals in a population, N , scattered over an area, A , will increase the lower the population density. It also follows that if individuals are clustered into independently moving co-residential groups, the average interaction distance between groups, I , will go as $I \propto \sqrt{A/N} (A/N)^{1/2}$, and so $I \propto D$. Therefore, the density-dependence of mobility not only minimizes the costs of mobility required to encounter resources, it also minimizes the interaction distance of individuals, promoting metapopulation connectivity.

We can make inferences from this model about the prehistoric hunter-gatherer world. Our model suggests a consistent relationship between mobility and population density, regardless of storage technology or environmental productivity. As such, while archaeological evidence of high levels of mobility, such as raw materials sourced to distant outcrops, may imply low population densities, the implications for archaeological evidence of storage in prehistoric hunter-gatherers still need to be made more explicit. Ethnohistorically, storage technologies correlate with sedentism, but this is because storage is related to the seasonality of resources. So, for example, increasing evidence of ceramic use throughout the Eurasian Upper Paleolithic thousands of years before farming seems to indicate important, and under-appreciated technological and culinary innovations in prehistoric hunter-gatherer societies (Bar-Yosef and Wang, 2012; Craig, 2021; Craig et al., 2013; Harry and Frink, 2009; Jordan and Zvelebil, 2009). However, the implications for mobility are complex as the increased use of ceramics may indicate technological innovations associated with increased use of seasonal resources after the Last Glacial Maximum as opposed to decreased emphasis on annual mobility budgets. Ethnohistorically, there are many examples of societies with high investments in storage technology that maintain high levels of mobility, such as groups in tundra and boreal forest environments. The ethnohistoric data that we have do not capture the full range of lifestyles and movement ecologies in the hunter-gatherer past. Those late Pleistocene/early Holocene hunter-gatherer populations associated with the onset of farming may have been atypical of other contemporary hunter-gatherer populations, and the ethnohistoric record may well not capture this part of the distribution.

In non-food producing economies, because food storage in itself does not increase the net productivity of an environment (i.e., kcal per unit area), it cannot reduce overall mobility costs as resources will always be depleted in patches requiring individuals to relocate to new patches which will then deplete requiring further relocation, and so on. Temporarily delaying the depletion of resources does not alter this fundamental energy budget. However, more productive environments will support more individuals per unit area because of faster resource supply rates (i.e., faster energy turnover). This means individuals can meet their energy demand in smaller home ranges in more productive environments, and so can expend less time and energy moving around landscapes as resources are easier to encounter. In addition, the average distance between people decreases in denser populations and so less energy needs to be expended in maintaining ties to the larger social network. Across all levels of food storage, denser populations tend to move less through the course of a year, at rates predicted by our theory.

The model we present here differs in important ways from other prominent models of density-dependent storage technology in the

literature. For example, Binford's (2001) population packing model treats population density as an exogenous variable to which populations respond through increasing storage technology and intensification as local resources become scarcer due to increased competition and mobility is reduced as there is less room in which to move; as a result, food storage and intensification are solutions to mounting exogenous costs. Essentially, this is a Platonic "necessity-is-the-mother-of-invention" model, in which case we would predict seeing food storage and intensification emerging where population densities and population growth are exceptionally high resulting in resource stress. However, we do not observe these patterns in ethnographic data, nor in the archaeological record of the earliest food producing economies of the late Paleolithic/early Holocene. Our model is explicitly ecological as population size is density-dependent and endogenous to the model; population density varies across different environments as more productive environments are, on average, capable of supporting greater population densities, as observed in data (Fig. 4B). Mobility reduction, or sedentism, in hunter-gatherer societies emerges endogenously from adaptations to particular environments.

Finally, our model suggests that food storage can only reduce total mobility costs in rare circumstances in hunter-gatherer societies. However, food storage is likely to lead to sedentism when coupled with technological inventions that increase the net productivity of environments (i.e., the density of harvestable resources per unit area), as this increases patch residence time while decreasing home range sizes. Intentionally increasing the net productivity of an environment requires either the artificial addition of nutrients, temperature, or water (the primary components of net primary productivity, thus increasing the resource supply rate) or the physical reconfiguration of resources on the landscape that changes the availability, predictability, diversity, and abundance of harvestable resources (Ritchie et al., in press). Artificially increasing the density of resources per unit area may be achieved through trade, mutualisms with food producers (Crowther et al., 2018; Gallois and Henry, 2021; Junker and Smith, 2017), domesticating wild resources, and ecosystem engineering (Hastings et al., 2007; Jones et al., 1997, 1994). While there is abundant evidence of hunter-gatherer niche construction (Smith, 2007; Sterelny, 2007; Thompson et al., 2021), including landscape burning (Bird et al., 2020), shell midden construction (Álvarez et al., 2011; Andrus, 2011; Smith, 2011), and seed dispersal (Deur and Turner, 2005; Lepofsky and Caldwell, 2013; Moss and Wellman, 2017), whether these have the effect of increasing net environmental resource supply rates is not clear. Nevertheless, storage does not necessarily imply sedentism: storage technologies increase the availability of food resources that would have otherwise depleted but does not increase the net productivity of environments.

CRediT authorship contribution statement

Marcus J. Hamilton: Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **B. Buchanan:** Writing – original draft, Writing – review & editing. **J. Lobo:** Conceptualization, Writing – review & editing. **R.S. Walker:** Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data and code are attached to the article

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jasrep.2024.104441>.

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